NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF ENGINEERING
DEPARTMENT OF ELECTRONIC ENGINEERING

Degree Programme Info
Of
Electronic Engineering

Electronic Engineering

Department of Electronic Engineering

Degree Programme Information

Programme:	Electronic Engineering
Programme Code:	TEE
Duration:	5 years
Minimum Credit Hours:	600
Maximum Credit Hours:	675
ZNQF Level:	SADC-QF-Level 8

Entry Requirements

- **Normal entry**
  Requires at three 'A' Level passes in Pure/Additional/Mechanical Mathematics, Physics and Chemistry.

- **Special entry requires:**

  a) National Diploma in Electronic Engineering or Telecommunication Engineering or Instrumentation and Control or Computer Engineering plus 2 years post National Diploma working experience.

  b) Higher National Diploma in Electronic Engineering or Telecommunication Engineering or Instrumentation and Control or Computer Engineering plus 1 year post Higher National Diploma working experience.

Learning Outcomes

- **Problem solving**
  Identify, formulate, analyse and solve complex engineering problems in electronic engineering, creatively and innovatively.

- **Application of scientific and engineering knowledge**
  Apply knowledge of mathematics, natural sciences, engineering fundamentals and an engineering speciality to solve complex engineering problems.

- **Engineering design**
  Perform creative, procedural and non-procedural design and synthesis of components, systems, engineering works.

- **Investigations, experiments and data analysis**
  Demonstrate competence to design and conduct investigations and experiments.

- **Engineering methods, skills and tools, including information technology**
  Demonstrate competence to use appropriate engineering methods, skills and tools, including those based on information technology.

- **Professional and technical communication**
  Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large.

- **Sustainability and impact of engineering activity**
  Demonstrate critical awareness of the sustainability and impact of engineering activity on the social, industrial and physical environment.

- **Individual, team and multidisciplinary working**
  Demonstrate competence to work effectively as an individual, in teams and in multidisciplinary environments.
- **Independent learning ability**
  Demonstrate competence to engage in independent learning through well-developed learning skills.
- **Engineering professionalism**
  Demonstrate critical awareness of the need to act professionally and ethically and to exercise judgment and take responsibility within own limits of competence.
- **Engineering management**
  Demonstrate knowledge and understanding of engineering management principles and economic decision making.
- To be entrepreneurial and create job in the related industry.
- Communicate well and clear, and show a good understanding of the discipline.

Programme Assessment

- **Coursework**
  Unless otherwise specified, continuous assessment shall contribute 25 % of the final marks.
- **Written Examination**
  Unless otherwise specified, the formal examination shall contribute 75 % of the final marks.

Programme Summary

Module Code	Module Title	Credits
	Part One	
PLC1101	Peace, Leadership and Conflict Transformation I	5
SCS1101	Introduction to Computer Science and Programming	10
SMA1116	Engineering Mathematics 1A	10
SPH1104	Modern Physics	10
TCE1103	Professional Engineering Skills I	5
TEE1103	Electrical Engineering - Basic Circuit Analysis	10
TEE1105	Electronic Engineering Workshop	5
TIE1101	Engineering Drawing I	10
PLC1201	Peace, Leadership and Conflict Transformation II	5
SCS1205	Software Engineering Concepts	10
SMA1216	Engineering Mathematics 1B	10
TEE1201	Graphics for Electronic Engineers	10
TEE1203	Electronic Engineering - Circuits and Devices	10
TEE1205	Electronics Engineering Workshop	5
TEE1211	Analogue Electronics	10
TEE1241	Electrical Measurements	10
	Part Two	
SMA2116	Engineering Mathematics II	10
TEE2101	Network Theory	10
TEE2102	Electrical Machines	10

TEE2103	Design and Project	5
TEE2104	Laboratory	5
TEE2111	Digital Electronics	10
TEE2112	Microprocessors	10
SMA2217	Engineering Mathematics III	10
TEE2201	Electromagnetic Fields and Materials	10
TEE2202	Electronic Drives	10
TEE2203	Design and Project	5
TEE2204	Laboratory	5
TEE2205	The Professional Engineer I	10
TEE2211	Digital Devices and Systems	10
TEE2232	Computer Engineering I	10
	Part Three	
SMA3116	Engineering Mathematics IV	10
TEE3101	Digital Signal Processing	10
TEE3103	Design and Project	5
TEE3104	Laboratory	5
TEE3111	Linear Integrated Circuits	10
TEE3121	Analogue Communication Engineering	10
TEE3132	Software Engineering	10
TEE3201	Electromagnetic Theory	10
TEE3203	Design and Project	5
TEE3204	Laboratory	5
TEE3221	Digital Communication Engineering	10
TEE3232	Embedded Systems	10
TEE3241	Control Engineering	10
	Part Four	
TEE4000	Industrial Attachment	120
	Part Five	
TEE5101	Engineering Management I	10
TEE5111	Integrated Circuit and Microelectronics	10
TEE5133	Communication Systems Performance	10
TEE5141	Modern Control Engineering	10
TEE5003	Honours Project	50
TEE5205	Engineering Management II	10
TEE5224	High Speed Networks	10
TEE5241	Industrial Control	10

Programme Activities Time Allocation

Programme Activity	Hours	Credits
Contact Hours		
Lectures	1621	162
Tutorials	336	34
Laboratory Work	450	45
Workshops	96	10
Industrial Attachment	1 Year	120
Assessment Hours		
Final written examinations	132	13
In-class tests	379	38

Practicals	421	42
Independent Study Hours		
Preparation for Assessments	403	40
Reading	458	46
Written assignments	326	33
Revisions	369	37
Maximum Credit for 80 % Modules Threshold		620

Peace, Leadership and Conflict Transformation I

Department of Business Management

Course Information

Course: Peace, Leadership and Conflict Transformation I
Course Code: PLC 1101
Credit Hours: 5

Lecturer Information

Name: David Foya
Email: david.foya@nust.ac.zw
Qualifications: Phd in Peace and Governance, Master in Peace and Governance, BA in Education

Course Synopsis

The Peace, Leadership and Conflict Transformation course is tailored in a manner to provide students with intellectual skills on the symbiotic relationship that exist on the three tier terms [peace, leadership and conflict]. The course attempts to probe into the interplay between these thematic motifs and show their role and complementarities in the process of human development. The course further seeks to provide a skills tool kit on how to analyze conflicts, identify their underlying causes, evaluate how conflict undermines the productive use of resources thereby plaguing development and how responsible leadership transforms adversity into peaceful, equitable and just global society in harmony with nature.
It is envisaged that the students who would successfully completed the course will be well grounded in the theory and practice to face the challenges of leadership and conflict at personal, community, national and global levels. The students would be able to trace the emerging patterns and conflict trends in Africa shall form the basis of reflection.

Learning Objectives

The objectives of this course are to provoke debate , through critical thinking, about conflict and conflict management, focusing on both endogenous and exogenous styles of leadership and conflict resolution approaches(negotiation, mediation, reconciliation etc). Therefore students should be able to:

- Describe and contextualise key theories and concepts of peace, leadership and conflict.
- Examine the origins and dynamism of both leadership and conflict.
- Understand the art of and functions of leadership in society and organisations and appreciate the interplay between leadership and conflict management.
- Develop descriptive and analytical research skills in leadership and conflict transformation.

Learning Outcomes

By the end of this course, participants will be able to:

- Demonstrate use of acquired skills by applying them to their context and able to transform the lives of the members of the community.
- Have a verbal or written demonstration of learnt concepts.

- Be able to explain the rationale for carrying out research in the three areas of peace, leadership and conflict transformation.
- Be able to identify research area, conduct research, gather data, analyse, interpret and make a write up.

Learning Requirements

- Students are expected to attend classes regularly. Attendance of classes is a requirement and excessive absence will result in failing the course.
- Written assignments should be typed, double spaced, (12 font size), with reference/bibliography at the end.
- Students are expected to participate in class discussions. Reading course materials in advance is essential for participation in discussion. Oral presentations shall include reviews, case studies from field visits, descriptive and analytical reading of the selected course materials.
- Written assignments should meet the university general requirements.
- Any written assignments should provide empirical description as evidence to substantiate on claims.

Topics

- Concept of Conflict and Peace
- Theories of Conflict and Peace
- Theories of Conflict
- Classical Social Structural Theories of Conflict
- Modern Structural Theories of Conflict
- Cultural, religious, Ethnic and Identity Based Conflicts
- Gender and Conflict
- Impact of Conflict on development
- Conflict Resolution Process

Assessment of Students

- Students are assessed through written assignments and tests that shall form part of course work assessment mark.
- Written assignments should meet University general requirements.
- Students are expected to participate in class discussion and to read course material.
- There shall be a three-hour final examination at the end of each semester.
- Marking of scripts rewarding of marks shall be based on:

    1. logic, analytical rigour and coherence.

2. understanding of the issues at hand.
3. creativity and critical engagement with the topic.
4. style, suave and presentation.
5. in the case of assignment, bibliography should be given.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (30 %), Final Exam (70 %)

Policies and Other Information

Key Text

- Aye, J (2000) Leadership in Contemporary Africa: Exploratory Study, New York United Nations.
- Banks, M. Four Conceptions of Peace: Unpublished Article.
- Deng , Z. (2002) Developmental Challenges in Africa. London, OUP.
- Galtung, J. (1997). Peace by Peaceful Means. London: Sage Press.
- Lederach, J.P. (1995). Preparing for Peace: Conflict Transformation Across Cultures. Syracuse: Syracuse University Press.
- www.beyondintractability.org
- www.berghof-handbook.net

Introduction to Computer Science and Programming

Department of Computer Science

Course Information

Course:	Introduction to Computer Science and Programming
Course Code:	SCS 1101
Credit Hours:	10

Lecturer Information

Name:	Daniel Musundire
Email:	daniel.musundire@nust.ac.zw
Qualifications:	MSc, (Hon) BSc in Computer Science

Course Synopsis

Information and Knowledge Societies, Evolution of Computers, Computer Organisation and Architecture: CPU; Memory; I/O, Number Systems and Conversions (Bin; Dec; Hex; Oct), Concepts of Computer Languages: highlevel languages; compiler; interpreter, Programming Techniques: grammar; recursion; Variables; Data types; Initialization; Comments; Keywords; Constants; Assignment, Programming constructs: branching; looping; recursion; Programming using data structures: arrays; lists; trees; hash tables; queues; stacks; files, Programming Algorithms for Problem Solving: Sorting; compression; numerical and encryption, Fundamentals Operating System, Fundamentals Data Bases, Fundamentals of Networks

Learning Objectives

By the end of the course, students will be able:

- To identify and describe the functions of computer hardware and computer systems.
- To understand the basics and administration of various operating systems including DOS, Windows, and Unix/Linux.
- To explain the structure, operation, programming, and applications of computers.
- To understand maintenance and diagnostics for computers.

Topics

- Introduction to computer hardware
- Components of a computer
- Introduction to operating systems
- Introduction to programming with C
- Computer software packages
- Computer maintenance and diagnostics

Assessment of Students

- Students are assessed through written assignments, practical assignments, in-class written tests and in-class practicals tests.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Operating Systems Design and Implementation, 3rd Edition, 2006 by A. Tanenbaum, A. Woodhull
- Modern Operating Systems, 2nd Edition, 2001 by A. Tanenbaum
- Software Architecture in Practice, 2nd Edition by L. Bass, P. Clememnts, R. Kazman
- Computer Science, 2nd Edition by P. Bishop

Engineering Mathematics 1A

Department of Applied Mathematics

Course Information

Course: Engineering Mathematics 1A
Course Code: SMA 1116
Credit Hours: 10

Lecturer Information

Name: Tinashe Gashirai
Email: tinashe.gashirai@nust.ac.zw
Qualifications: MSc in Applied Mathematical, (Hon) BSc in Applied Mathematics

Course Synopsis

The module explores calculus in one variable i.e. Limits and continuity of functions, Differentiation, Leibniz's Rule, L'Hopitals Rule, Elementary functions including hyperbolic functions and their inverses; Integration – techniques including reduction formulae; Applications – arc-length, area, volumes, moments of inertia, centroids; Plane polar coordinates; Complex Numbers: Basic algebra; De Moivre's theorem; Complex exponentials; Linear Algebra: Vector algebra in 2 and 3 dimensions; Scalar and vector products; Equations of lines and planes.

Topics

- Calculus in One Variable
  - Limits and Continuity of functions
  - Differentiation
  - Leibniz's rule, L'Hopital's Rule
  - Elementary functions including hyperbolic function and their inverse
  - Integration techniques including reduction formulae
  - Application of integration
    arc length, volumes, moments of inertia, centroids, plane polar coordinates
- Complex Numbers
  - Basic algebra
  - De Moivre's theorem
  - Complex exponentials
- Linear Algebra
  - Vector algebra in two and three dimensions
  - Scalar and Vector products
  - Equations of line and planes

Assessment of Students

- Students are assessed through:

  1. course work consisting of two in-class tests and two assignments.
  2. end of semester final examination.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Calculus on a single variable, 8th Edition, 2004 by L. Hostetler
- Calculus, 6th Edition, 2003 by J. Stewart
- Calculus: Concepts and Connections, 3rd Edition, 1998 by R. T. Smith, R. B. Minton
- Calculus: Early Transcendental Functions, 3rd Edition, 1998 by R. T. Smith, R. B. Minton
- Calculus, 8th Edition, 2005 by H. Anton, I. Bivens, S. Davis

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Professional Engineering Skills I

Department of Chemical Engineering

Course Information

Course:	Professional Engineering Skills I
Course Code:	TCE 1103
Credit Hours:	10

Lecturer Information

Name:	Nonhlanhla G. Mguni
Email:	nonhlanhla.mguni@nust.ac.zw
Qualifications:	MSc in Chemical Engineering, (Hons) BEng in Chemical Engineering

Course Synopsis

The course covers study methods, communication principles, technical definitions, descriptions and instructions. It delves into tables and graphs, letters, memoranda and curricula vitae and written reports. It also covers Word processing and computer jargon, interview techniques, running a meeting, reading, understanding and summarising technical articles.

Topics

- A Guide to Writing as an Engineer
- Business Correspondence
- Communication Principles
- Curriculum Vitae
- Effective Study Habits and Methods
- Interview Skills
- Presentations
- Writing Engineering Reports and Writing Common Engineering Documents

Assessment of Students

The assessment of students comprises:

- a continuous assessment of written in-class tests.
- an examination written at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Course Information

Course:	Electrical Engineering - Basic Circuit Analysis
Course Code:	TEE 1103
Credit Hours:	10

Lecturer Information

Name:	Svetlana A. Bebova
Email:	swetlana.bebova@nust.ac.zw
Qualifications:	MSc in Radio Electronic Engineering

Course Synopsis

The course introduces the student to general concepts of current, voltage, resistance and circuits (dc and ac). It also covers elements of loop and nodal analysis, basic networks and theorems, Delta-Wye conversions and network theorems. It delves into Capacitor and inductor circuits, Transient analysis, DC and AC power. It also delves into forced response and sinusoidal steady-state response.

Topics

- Introduction and Review of Electricity
- Basics of Circuit Analysis (DC)
- Methods of Circuit Analysis (DC)
- Network Theorem (DC)
- Capacitance
- Magnetic Circuits
- Inductance
- Alternating Current
- Analysis of AC Circuits
- Application of Network Theorems to AC Circuits
- AC Power

Assessment of Students

- Students are assessed through written assignments and written in-class tests that shall form part of continuous assessment mark.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Introductory Circuit Analysis, 11th Edition, 2007 by R. L. Boylestad
- Introductory Electronic Devices and Circuits: Electron Flow Version, 7th Edition, 2006 by R. Paynter
- Electronic Measurement and Instrumentation, 1996 by K. B. Klaassen, S. Gee

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Course Information

Course:	Electronic Engineering Workshop
Course Code:	TEE 1105
Credit Hours:	5

Course Synopsis

Identification of electronic components; soldering on veroboards; production of PCBs; fabrication of circuit boards and interconnection; relevant mechanical construction.

Engineering Drawing I

Department of Industrial and Manufacturing Engineering

Course Information

Course: Engineering Drawing I
Course Code: TIE 1101
Credit Hours: 10

Lecturer Information

Name: Noel M. Dewa
Email: noel.dewa@nust.ac.zw
Qualifications: BTech in Mechanical Engineering

Course Synopsis

The module examines an introduction into plane geometry, space geometry, first and third angle projection. The module delves into dimensioning, pictorial views, freehand sketching, drawing of common objects, sectioning and intersections. It also covers developments, conventions and assembly drawings.

Topics

Assessment of Students

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Peace, Leadership and Conflict Transformation II

Department of Business Management

Course Information

Course:	Peace, Leadership and Conflict Transformation II
Course Code:	PLC 1201
Credit Hours:	5

Lecturer Information

Name:	Octavious Masunda
Email:	octavious.masunda@nust.ac.zw
Qualifications:	Master in Peace and Governance, BA in Education

Course Synopsis

The Peace, Leadership and Conflict Transformation course is tailored in a manner to provide students with intellectual skills on the symbiotic relationship that exist on the three tier terms [peace, leadership and conflict]. The course attempts to probe into the interplay between these thematic motifs and show their role and complementarities in the process of human development. The course further seeks to provide a skills tool kit on how to analyze conflicts, identify their underlying causes, evaluate how conflict undermines the productive use of resources thereby plaguing development and how responsible leadership transforms adversity into peaceful, equitable and just global society in harmony with nature.
It is envisaged that the students who would successfully completed the course will be well grounded in the theory and practice to face the challenges of leadership and conflict at personal, community, national and global levels. The students would be able to trace the emerging patterns and conflict trends in Africa shall form the basis of reflection.

Learning Objectives

The objectives of this course are to provoke debate , through critical thinking, about conflict and conflict management, focusing on both endogenous and exogenous styles of leadership and conflict resolution approaches(negotiation, mediation, reconciliation etc). Therefore students should be able to:

- Describe and contextualise key theories and concepts of peace, leadership and conflict.
- Examine the origins and dynamism of both leadership and conflict.
- Understand the art of and functions of leadership in society and organisations and appreciate the interplay between leadership and conflict management.
- Develop descriptive and analytical research skills in leadership and conflict transformation.

Learning Outcomes

By the end of this course, participants will be able to:

- Demonstrate use of acquired skills by applying them to their context and able to transform the lives of the members of the community.
- Have a verbal or written demonstration of learnt concepts.

- Be able to explain the rationale for carrying out research in the three areas of peace, leadership and conflict transformation.
- Be able to identify research area, conduct research, gather data, analyse, interpret and make a write up.

Learning Requirements

- Students are expected to attend classes regularly. Attendance of classes is a requirement and excessive absence will result in failing the course.
- Written assignments should be typed, double spaced, (12 font size), with reference/bibliography at the end.
- Students are expected to participate in class discussions. Reading course materials in advance is essential for participation in discussion. Oral presentations shall include reviews, case studies from field visits, descriptive and analytical reading of the selected course materials.
- Written assignments should meet the university general requirements.
- Any written assignments should provide empirical description as evidence to substantiate on claims.

Topics

- Leadership
- Leadership and Conflict Transformation Skills
- Non-Violence
- Classification of Leaders
- Nexus Between Leadership and Conflict
- Interplay: Leadership, Conflict and Development

Assessment of Students

- Students are assessed through written assignments and tests that shall form part of course work assessment mark.
- Written assignments should meet University general requirements.
- Students are expected to participate in class discussion and to read course material.
- There shall be a three-hour final examination at the end of each semester.
- Marking of scripts rewarding of marks shall be based on:
  1. logic, analytical rigour and coherence.
  2. understanding of the issues at hand.
  3. creativity and critical engagement with the topic.
  4. style, suave and presentation.
  5. in the case of assignment, bibliography should be given.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (30 %), Final Exam (70 %)

Policies and Other Information

Key Text

- Aye, J (2000) Leadership in Contemporary Africa: Exploratory Study, New York United Nations.
- Banks, M. Four Conceptions of Peace: Unpublished Article.
- CARTWRIGHT, J (2004) Political Leadership in Africa, London.
- Clocker, C. et al (1994) Multiparty Mediation The Conflict Cycle.
- Deng , Z. (2002) Developmental Challenges in Africa. London, OUP.
- Mazrui, Ali ((1990) Cultural Forces in World Politics, Nairobi, James Currey Ltd.
- McDonald, J (1996) Multi-Track Diplomacy: A System Approach to Peace 3rd Edition, Kumarian Press.
- Sandole, D. J. D and Merwe van der Hugo. (1993). Conflict Resolution Theory and Practice: Integration and Application Manchester: Manchester University Press.
- Galtung, J. (1997). Peace by Peaceful Means. London: Sage Press.
- Lederach, J.P. (1995). Preparing for Peace: Conflict Transformation Across Cultures. Syracuse: Syracuse University Press.

Software Engineering Concepts

Department of Computer Science

Course Information

Course:	Software Engineering Concepts
Course Code:	SCS 1205
Credit Hours:	10

Lecturer Information

Name:	Khulekani Sibanda
Email:	khulekani.sibanda@nust.ac.zw
Qualifications:	MSc, (Hon) BSc in Computer Science

Course Synopsis

The module has software development life cycle; Requirements, specification, design implementation and testing, coding, maintenance; Function-oriented design methodologies; Documentation; Implementation strategies; Debugging, anti-bugging; Introduction to specifications, verification and validation; Elementary proof of correctness; Code and design reading; Structured walkthroughs; Testing strategies; Software reliability issues; Configuration Management; CASE tools; Programming languages; Compilers; The DotNet framework and programming in C.

Topics

- Introduction
- System Engineering
- Software Processes
- Project Management
- Software Requirements
- Requirements Engineering Processes
- System Models
- Software Prototyping
- Formal Specifications
- Architectural Design
- Distributed Systems Architectures
- Object-Oriented Design
- Real-time Software Design
- Design with Reuse
- User Interface Design
- Dependability
- Dependable Systems Specification

- Dependable Software Development
- Verification and Validation
- Defect Testing
- Critical Systems Validation
- Managing People
- Software Cost Estimation
- Quality Management
- Process Improvement
- Legacy Systems
- Software Change
- Software Re-engineering
- Configuration management

Assessment of Students

- Students are assessed through written assignments and tests that shall form part of course work assessment mark.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Software Engineering, 9th Edition by I. Sommerville

Engineering Mathematics 1B

Department of Applied Mathematics

Course Information

Course:	Engineering Mathematics 1B
Course Code:	SMA 1216
Credit Hours:	10

Lecturer Information

Name:	Tinashe Gashirai
Email:	tinashe.gashirai@nust.ac.zw
Qualifications:	MSc in Applied Mathematical, (Hon) BSc in Applied Mathematics

Course Synopsis

This course aims at providing the student with the basic concepts of Functions of Several Variables, Matrices and Ordinary Differential Equations which are motivated by Radiography and Engineering Sciences applications while emphasizing the informed use of modern numerical analysis methods. The objective of this course, therefore, is to present mathematical theories that Engineering Scientists have successfully used so far in a manner that highlights the historical and intellectual connections between the applications in Mathematics, Physics and Engineering.

Learning Requirements

Students are expected to have completed and passed the following courses:

- SMA 1116

Topics

- Functions of Several Variables
  - Limits and Continuity
  - Partial Derivatives and the Chain rule
  - Tangent planes and normal lines
  - Extrema and the Lagrange Multipliers
  - Applications of Functions of Several Variables
- Matrices (Linear Algebra)
  - Basic Operations of Matrices
  - Properties and definitions
  - Elementary Row Operations (Gauss Elimination)
  - Systems of Equations
  - Determinants and inverses
  - Eigenvalues and eigenvectors
  - Linear independence
  - Application of Matrices
- Ordinary Differentail Equations

- First Order Differential Equations
- Variable separable
- Integrating factor
- Exact equations
- Homogeneous ODEs
- Linear second ODEs with constant coefficients
- Variation of Parameters
- System of Equations
- Applications of ODEs to Physics and Engineering

Assessment of Students

Two in-class tests will constitute the major assessment from which a continuous assessment mark will be obtained. There will be final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

The following textbooks are recommended for studying in this course. However, any other textbook with content relevant to the course is appropriate for personal enrichment.

- Elementary Calculus, 2010 by H. J. Keisler
- Calculus, 2007 by J. Stewart
- A first course in Differential Equations with Modelling Application by D. G. Zill
- Calculus and its Applications, 10th Edition, M. L. Bitternger, D. J. Ellenbogen
- Linear Algebra with Applications, 3rd Edition, 2004 by Bretscher

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Graphics for Electronic Engineers

Department of Electronic Engineering

Course Information

Course:	Graphics for Electronic Engineers
Course Code:	TEE 1201
Credit Hours:	10

Lecturer Information

Name:	Bhekisisa Nyoni
Email:	bhekisisa.dube@nust.ac.zw
Qualifications:	MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

The module focuses on graphical techniques for drawing circuit diagrams, logic circuits, flowcharts. It covers concepts of engineering drawings, presentation of graphs and design of artwork for printed circuit boards. Use of pictures and cartoons and use of CIRCUIT MAKER PRO program for graphical design.

Topics

- Concepts of Engineering Drawing
- Graphical Techniques for Drawing
  circuit diagrams, logic circuits, flowcharts
- Presentation of graphs
- Design of artwork for printed circuit boards
- Use of pictures and cartoons
- Use of Circuit Maker Pro programs for graphical design
- Use of Proteus programs for graphical design

Assessment of Students

- Students are assessed through:
  1. course work consisting of class work, assignments and class tests.
  2. end of semester final laboratory examination with practical components on the use of circuit design software and/or other graphical communication tools.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Introduction to Graphics Communications for Engineers, 4th Edition , 2013 by G.B. Bertoline
- Presentation graphics for engineering, science and business, 2005, CRC Press by P.H. Milne

Course Information

Course: Electronic Engineering - Circuits and Devices
Course Code: TEE 1203
Credit Hours: 10

Lecturer Information

Name: Svetlana A. Bebova
Email: swetlana.bebova@nust.ac.zw
Qualifications: MSc in Radio Electronic Engineering

Course Synopsis

The module outlines rectifying and Zener diodes: structure, operation, characteristics and parameters; Diode applications: rectifiers and power supplies; clippers and clamps; Schottky diode; Transistors; Bipolar Junction Transistors (BJTs): structure, terminals and operation; BJT configurations, static characteristics and parameters; biasing methods; d;c; circuit analysis and design; Darlington pair; BJT packages and data sheet; Field Effect Transistors (FETs): types, structure, terminals and operation; Configurations and static characteristics; d;c; circuit analysis; Power devices and heat sinks; Optoelectronic and photo- electronic devices: Light-Emitting Diodes (LEDs), infra-red diodes, 7-segment displays, Liquid Crystal Displays (LCDs), photodiode and phototransistor; Applications; Thermistors: structure, types, operation and applications.

Topics

- Semiconductor Diodes
- Semiconductor Materials and BJT
- Diodes Application
- Bipolar Junction Transistor
- BJT Biasing and equivalent circuits
- Field Effect Transistors

Assessment of Students

- Students are assessed through written assignments and written in-class tests that shall form part of course work assessment mark.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Electronic Principles, 8th Edition, 2015 by A. Malvino, D. Bates
- The Art of Electronics, 3rd Edition, 2015 by P. Horowitz

Electronic Engineering Workshop

Department of Electronic Engineering

Course Information

Course: Electronic Engineering Workshop
Course Code: TEE 1205
Credit Hours: 5

Course Synopsis

Reading electronic circuit diagrams. Recognition of standard electrical and electronic symbols. Electronic circuits board design, fabrication technology and practice. Design and fabrication of metal cabinets for projects. Placement of components on the printed circuit board. Soldering practice. Placement of project in the cabinet. Finishing and labelling of electronic equipment. Interpretation of electrical specifications for equipment.

Analogue Electronics

Department of Electronic Engineering

Course Information

Course:	Analogue Electronics
Course Code:	TEE 1211
Credit Hours:	10

Lecturer Information

Name:	Zedekia Madumbu Nyathi
Email:	zedekia.madumbu.nyathi@nust.ac.zw
Qualifications:	MSc in Electronic Engineering, FETC

Course Synopsis

An introduction to stabilised power supplies, small-signal models of differential, single stage, multi-stage and integrated circuit amplifiers, oscillators, wave shaping and switching circuits. Covers high frequency effects and stability and performance measurement. Devles into electronic test equipment and verification of principles.

Topics

- Field Effect Transistor
- Biasing of BJTs
- Biasing of FETs and MOSFETs
- Small Signal Analysis of BJTs
- Small Signal Analysis of FETs
- High Frequency Response of FETs and BJTs
- Feedback and Oscillators
- Operational Amplifiers
- Power Amplifiers

Assessment of Students

- Students are assessed through two written in-class tests which shall form the continuous assessment mark.
- There shall be final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Electronic Engineering, 1st Edition, 2008 by A. P. Godse, U. A. Bakshi
- Understanding Electronics, 3rd Edition, R. H. Warring, G. R. Slone
- Electronic Devices and Circuit Theory, 7th Edition by R. Boylestad, L. Nashelsky
- The Illustrated Dictionary of Electronics, 8th Edition

Electrical Measurements

Department of Electronic Engineering

Course Information

Course:	Electrical Measurements
Course Code:	TEE 1241
Credit Hours:	10

Lecturer Information

Name:	Bhekisisa Nyoni
Email:	bhekisisa.dube@nust.ac.zw
Qualifications:	MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

This course delves into basic measuring devices, ammeters, voltmeters, transducers, accuracy, precision and components. It looks into electronic measuring instruments, digital voltmeters, multimeters. It also covers oscilloscopes as measurement instruments, measurement of AC power and measurement of nonelectrical parameters.

Topics

- Fundamentals of Measurement System
- Sensors
- Signal Conditioning
- Analogue Instruments
- Digital Instruments
- Frequency and Waveform Measurement
- Recording Instruments
- Telemetry
- Automatic Measurement Systems
- Errors in Measurement

Assessment of Students

- Students are assessed through written assignments and written in-class tests that shall form part of course work assessment mark.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Engineering Mathematics II

Department of Applied Mathematics

Course Information

Course:	Engineering Mathematics II
Course Code:	SMA 2116
Credit Hours:	10

Lecturer Information

Name:	Mlamuli Dhlamini
Email:	mlamuli.dhlamini@nust.ac.zw
Qualifications:	PhD in Applied Mathematics, MSc in Industrial Mathematics, BSc in Applied Mathematics

Learning Aim

This course aims at introducing students to dealing with functions of more than one variable and Fourier series.

Learning Objectives

At the end of the course students must be able to:

- evaluate multiple integrals.
- change from one co-ordinate system to another.
- use the integral theorems.
- to evaluate the fourier series for a given function.

Learning Requirements

Students are expected to have completed and passed the following courses:

- SMA 1116
- SMA 1216

Topics

- Multiple Integrals
- Vector Calculus
- Vector Integral Function
- Integral Theorem
- Fourier's Series

Assessment of Students

At least two in-class tests will be given as a continuous assessment during the semester before a final exam at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment tests (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Calculus with analytic geometry by Dennis Zill
- Vector Analysis Schaum's outline Series by Murray R Spiegel
- Vector calculus by Miroslav Lovric
- Elementary applied Partial differential equations by Richard Haberman
- Introduction to Fourier Analysis by Norman Morrison

Network Theory

Department of Electronic Engineering

Course Information

Course: Network Theory
Course Code: TEE 2101
Credit Hours: 10

Lecturer Information

Name: Magripa Nleya
Email: magripa.nleya@nust.ac.zw
Qualifications: MSc in Telecommunication Engineering

Learning Aim

To instill concepts of Network Theory into students.

Learning Objectives

By the end of the course, candidates should be able to:

- analyse electrical circuits.
- identify the type of the circuit and solve problems using appropriate method for the given circuit.

Topics

- DC circuits analysis
  - First order circuits
  - Second order circuits
- Analysing AC circuits analysis
  - Sinusoids and phasors
  - Sinusoidal steady-state analysis
  - Frequency response
- Advanced circuit analysis
  - The Laplace Transform
  - The Fourier series
  - Fourier Transform
  - Two-port networks

Assessment of Students

Tests, Laboratory, Assignments and Final Examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Introductory circuit analysis by Boylestard
- Fundamentals of electric circuits by C.K. Alexander, M.N.O. Sadiku.
- Lecture notes.

Design Project

Department of Electronic Engineering

Course Information

Course: Design Project
Course Code: TEE 2103
Credit Hours: 5

Course Synopsis

This course consists of a number of designs, constructs; Test type projects carried out in the laboratories and computer room.

Digital Electronics

Department of Electronic Engineering

Course Information

Course: Digital Electronics
Course Code: TEE 2111
Credit Hours: 10

Lecturer Information

Name: Svetlana A. Bebova
Email: swetlana.bebova@nust.ac.zw
Qualifications: MSc in Radio Electronic Engineering

Course Synopsis

This module looks at numerical systems i.e. binary, octal, hexadecimal and it also looks at system conversions, mathematical operations in straight and BCD code, logic gates, truth tables, boolean algebra theorems and K-maps and Minimization of logic expressions. It also delves into combinational logic applications and design i.e. arithmetic circuits, encoders and decoders, code converters, multiplexers and de-multiplexers as well as Flip-Flops

Topics

- Numerical Systems
- Logic Gates
- Combinational Logic
- Combinational Logic Applications
- Flip-flops

Assessment of Students

- Continuous assessment will consist of tests and assignments.
- Final assessment will consist of an examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Fundamentals of Digital Electronics, 1998 by B. Paton
- Digital Systems Principles and Applications, 10th Edition by R. J. Tocci, N. S. Widmer, G. L. Moss
- Theory and Design of Electronic Circuits by E. Tait

Microprocessors

Department of Electronic Engineering

Course Information

Course: Microprocessors
Course Code: TEE 2112
Credit Hours: 10

Lecturer Information

Name: Reginald Gonye
Email: reginald.gonye@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Learning Aim

To introduce students to microprocessors and microprocessor based systems (computers).

Learning Objectives

By the end of the course, candidates should be able to:

- Understand what a Microprocessor is and how it works.
- Appreciate the necessary components and understand the required hardware connections between the components that make up a microprocessor-based system or "computer" as it is known.
- Understand microprocessor instructions and their representation using assembly language statements.
- Design programs to solve novel situations using microprocessors.
- Appreciate the operation of microprocessor-based systems.

Topics

- Introduction
- Data Representation
- Main Memory for Microprocessor
- Input/Output for a Microprocessor
- Microprocessor Internal Architecture
- Microprocessor Instructions
- Designing Microprocessor Programs
- Stack, Subroutines and Interrupts
- Control of Peripherals
- Internal Operation of Microprocessors
- Application of Microprocessors and Performance

Assessment of Students

- Continuous assessment will consist of tests, assignments and laboratory work.
- Final assessment will consist of an examination at the end of the semester. The examination shall comprise of three sections as follows:

    a) SECTION A – answer all questions; mainly factual recall; total of 40 marks
    b) SECTION B – 3 questions each with 15 marks; answer any two; questions will examine at a level of candidates' understanding of a microprocessor (understand what programs do, transforming basic operations into programs, signal sequences as instructions execute, etc); total of 30 marks
    c) SECTION C – 3 questions each with 30 marks; answer any one; questions will examine the candidate's ability to design a microprocessor-based solution to a novel situation; total of 30 marks.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Microprocessor Theory and Applications with 68000/68020 and Pentium, 2008 by R. Mohamed
- Microprocessors and Microprocessor-based System Design, 2nd Edition, 1995 by R. Mohamed
- Introduction to Microprocessors, 3rd Edition, 1989 by A. P. Mathur
- The Z80 manual
- Microprocessors: Essentials, Components and Systems by R. Meadows, A. J. Parsons

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Engineering Mathematics III

Department of Applied Mathematics

Course Information

Course: Engineering Mathematics III
Course Code: SMA 2217
Credit Hours: 10

Lecturer Information

Name: Mlamuli Dhlamini
Email: mlamuli.dhlamini@nust.ac.zw
Qualifications: PhD in Applied Mathematics, MSc in Industrial Mathematics, BSc in Applied Mathematics

Course Synopsis

The course covers laplace transforms i.e. definitions, basic ideas and applications to ordinary differential equations. It also delves into statistics i.e. an introduction to applied statistics, introduction to probability and distribution theory, descriptive statistics/initial data exploration, summary statistics and graphical presentation of data. I also covers point estimation/test of hypothesis, interval estimation, analysis of variance and Regression analysis

Learning Requirements

Students are expected to have completed and passed the following courses:

- SMA 1116
- SMA 1216
- SMA 2116

Topics

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- Data Analysis
- Probability
- Random Variables and Discrete Distribution
- Continuous Probability Distribution

- Sampling Distribution
- Point and Interval Estimation
- Large Sample Estimation
- Large-Sample Tests of Hypothesis
- Inference From Small Sample
- The Analysis of Variance
- Simple Linear Regression and Correlation
- Multiple Linear Regression

Assessment of Students

At least two in-class tests will be given as a continuous assessment during the semester before a final exam at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment tests (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Laplace Transforms, 1973 by K.A. Stroud
- Differential Equations: Higher Order Differential Equations, Mathematics Notes by P. Dawkins
- Higher Engineering Mathematics, 5th Edition by J. Bird
- Mathematics, Mechanic and Probability, 1984 by L. Bostock, S. Chandler

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Electromagnetic Fields and Materials

Department of Electronic Engineering

Course Information

Course: Electromagnetic Fields and Materials
Course Code: TEE 2201
Credit Hours: 10

Lecturer Information

Name: Svetlana A. Bebova
Email: swetlana.bebova@nust.ac.zw
Qualifications: MSc in Radio Electronic Engineering

Course Synopsis

The module focuses on semiconductors materials i.e. atomic theory and band-gap. It also focuses on the p-n junction characteristics i.e. field theory, electric and magnetic fields.

Topics

- Classification Engineering Materials
- Conductors
  - The Structure of an atom
  - Atomic Arrangement
  - Electrical Conductivity
  - Band Theory and Temperature effects
  - Applications
- Semiconductors
  - Energy Gaps
  - Intrinsic Semiconductors
  - Extrinsic Semiconductors
  - Contact Phenomenon i.eṖ-N Junction and Contact Metal-semiconductor
  - Applications
- Dielectric Material
  - Polarisation
  - Dielectric constant
  - Dielectric Strength
  - Frequency and Effects of Temperature
- Magnetic Material
  - Magnetic Dipoles and Magnetic Moments
  - Magnetisation and permeability
  - Interactions between Magnetic Dipoles and Magnetic Field

- Domain Structure
- Applications

Assessment of Students

- Students are assessed through written assignments and written in-class tests that shall form part of course work assessment mark.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Electronic Drives

Department of Electronic Engineering

Course Information

Course: Electronic Drives
Course Code: TEE 2202
Credit Hours: 10

Lecturer Information

Name: Busiso Mtunzi
Email: busiso.mtunzi@nust.ac.zw
Qualifications: PhD in Physics (Renewable Energy), PhD in Physics (PV/T Hybrid Systems), MSc in Renewable Energy, BTech in Applied Physics

Course Synopsis

Power Electronics: Characteristics drive requirements and device protection i.e. Converters: DC-DC, DC-AC, AC-AC, AC-DC and control techniques. Power and distortion factor. Application of AC and DC motors. Motor Control i.e. variable speed drives, regenerative braking, slip energy recovery, four quadrant operations. Selection and sizing of motor drive systems. Transducers for power electronics application.

Learning Objectives

The course should enable you to:

- Understand the operation of power electronic devices in electrical energy control.
- Understand the application of power electronic devices in the lower and upper power end applications of power electronic devices:
  - Lower Power End
    Switched Mode Regulated Power supplies.
  - Upper Power End
    These include diverse industrial applications in power electronic converters for variable-speed drives, in steel rolling, textile, paper rolling mills, machine tools and other automation drives.

Learning Outcomes

At the conclusion of the course, the student is expected to:

- Have a basic understanding of modern power semiconductor devices; their strength, and their switching and protection techniques. These include power diodes, bipolar and MOSFET power transistors, silicon controlled rectifiers (SCRs) / thyristors , insulated-gate bipolar transistors (IGBT) and gate turn-off thyristors (GTO).
- Understand the operation and develop analysis skills of DC-DC; DC-AC; AC-DC and AC-AC converters.
- Understand and analyze the qualities of waveforms at input and output ends of these converters (RF; FF; DF).

Topics

- Introduction
  Overview of power semiconductor devices, characteristics.
- Diode (Uncontrolled) rectifiers and Harmonic and Distortion factors
- Controlled AC-DC rectifiers on Resistive and Inductive loads
- Choppers, DC-DC converters, Control issues
- DC-AC Converters (Inverters)
- Gate drive and Snubber circuits.
- Device losses and thermal design.

Assessment of Students

There will be two written assignments and two in-class written test (Test 1 on Devices's operation and protection, Test 2 on SCR and Choppers). At the end of the semester, there shall be a final written examination.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Power Electronics Circuits, Devices and Applications, 3rd edition by M.H. Rashid
- Principles of Electric Machines and Power Electronics, 1989 by P. C. Sen

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The Professional Engineer I

Department of Electronic Engineering

Course Information

Course:	The Professional Engineer I
Course Code:	TEE 2205
Credit Hours:	10

Lecturer Information

Name:	Fidelis N. Mugarisanwa
Email:	fidelis.mugarisanwa@nust.ac.zw
Qualifications:	MPhil in Electronic Engineering, BSc in Computer Science

Course Synopsis

The course aims at promoting the student's understanding of his/her role in society as the future engineer. At the end of the course, the student should be able to at least appreciate his or her duty as a potential innovator and/or service provider who must act within the moral-legal framework. The student must also be able to link every engineering activity with the needs of society and be conscious of everyday problems that may challenge their ability to generate solutions to some of those problems. Every possible attempt will be made to encourage the student to keep abreast with ever-changing technology and the subsequent demands imposed on him or her from the community that they live in. The students should also be able to independently manage or at least develop interest in small-scale multidisciplinary projects.

Topics

- Introduction
- Idea Generation and Product Development
- Engineering Business Environment
- Business Entity Types
- Business Models
- Research Methods
- Project Management
- Ethics in Engineering
- Safety in the Workplace

Assessment of Students

- Continuous assessment will consist of in-class presentations, written assignments and written in-class tests.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Digital Devices and Systems

Department of Electronic Engineering

Course Information

Course: Digital Devices and Systems
Course Code: TEE 2211
Credit Hours: 10

Lecturer Information

Name: Svetlana A. Bebova
Email: swetlana.bebova@nust.ac.zw
Qualifications: MSc in Radio Electronic Engineering

Course Synopsis

The module focuses on Flip-Flops review i.e. master-slave Flip-Flops, Shift registers. Counters i.e asynchronous with mod(numbers) $< 2N$, synchronous, down counters, up/down counters, integrated circuits counters. Memory devices i.e. registers, RAM, ROM and PROM. Combinational circuits i.e. adders, multiplexer, decoders and code coverters. Arithmetic circuits, arithmetic logic unit. Design of combinational circuits using MSI devices. Sequential system designs. Design of synchronous sequential systems.

Topics

- Memory devices
- ROM
- RAM
- PROM
- PLA
- Combinational circuits
  - Adders
  - Multiplexers
  - Decoders
  - Code converters
- Design of combinational circuits using MSI devices
- Arithmetic Logic Unit (ALU)
- Sequential system designs
- Design of asynchronous sequential systems

Assessment of Students

- Students are assessed through written assignments and written in-class tests that shall form part of course work assessment mark.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Digital systems: Principles and applications, 1985 by R. J.Tocci
- Digital Fundamentals, 1990 by T. L. Floyd
- Logic and computer design fundamentals, 1997 by M. M. Mano, C. R.Kime
- Electronic logic systems, 1994 by A.E.A. Almain
- Digital logic and computer design, 1979 by M. M. Mano
- Introduction to Digital Logic Design, 1994 by J. P. Hayes

Computer Engineering I

Department of Electronic Engineering

Course Information

Course: Computer Engineering I
Course Code: TEE 2232
Credit Hours: 10

Lecturer Information

Name: Lovemore Gunda
Email: loveugunda@gmail.com
Qualifications: MEng in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

The course provides an in-depth understanding of software engineering principles with a focus on object-oriented programming (OOP). The module covers fundamental OOP concepts, such as encapsulation, inheritance, polymorphism and data abstraction, through practical C++ examples. Students learn how to design and implement classes, manage memory and use pointers and virtual functions for dynamic behavior. This course also equips students with skills to develop modular, reusable and efficient software solutions.

Topics

- Introduction to Software Engineering
- Object oriented programming in C++
- Data abstraction and encapsulation
- Polymorphism
- Inheritance
- Pointers and Objects
- Association relationships
- Arrays
- Pointers
- Strings
- OOP using C++

Assessment of Students

Students are assessed through:

1. written assignments, practical assignments, written in-class test and practical in-class test.
2. end of semester final examination

Course Evaluation and Grading Policies

Grades are based on: Continuous Assessment (25 %), Final Exam (75 %)

Engineering Mathematics IV

Department of Applied Mathematics

Course Information

Course: Engineering Mathematics IV
Course Code: SMA 3116
Credit Hours: 10

Lecturer Information

Name: Hausitoe Nare
Email: hausitoe.nare@nust.ac.zw
Qualifications: (Hon) BSc in Applied Mathematics, MSc in Operations Research

Course Synopsis

This course introduces engineering students to partial differential equations, their solution using the method of separation of variables, to the power series solution of ordinary differential equations and to special functions, and to numerical methods for equation solution, curve fitting, differentiation, integration and the solution of ordinary differential equations.

Learning Objectives

On completion of the course students should be able to:

- Solve simple partial differential equations by direct integration or by the use of the techniques of ordinary differential equations.
- Solve eigenvalue and eigenfunction problems that arise in simple boundary value problems.
- Use the method of separation of variables to solve boundary value problems with the Laplace, Diffusion and Wave equations.
- Use the power series method to solve ordinary differential equations in the neighbourhood of an ordinary point.
- Identify and classify the singular points of an ordinary differential equation.
- Use the method of Frobenius to solve ordinary differential equations in the neighbourhood of an regular singular point.
- Solve Bessel's equation and others to identify a solution as a particular special function.
- Understand the concept of different types of error encountered in numerical methods.
- Understand and use a variety of methods to fit a curve to a set of points.
- Understand and use a variety of methods, including the Three Point Method, to differentiate a function of one variable.
- nderstand and use a variety of methods, including Simpson's rule, to evaluate definite integrals.

Learning Requirements

Students are expected to have completed and passed the following courses:

- SMA 1116
- SMA 1216
- SMA 2116
- SMA 2217

Topics

- Ordinary Differential Equations
  - Power series solutions
  - Frobenius method
  - Special functions and their properties
  - Legendre polynomials
  - Bessel functions
- Partial Differential Equations
  - Solutions of partial differential equations
  - Method of separation of variables
- Numerical Methods
  - Errors, relative and absolute
  - The solution of non-linear equations
  - The solution of linear systems
  - Interpolation and polynomial approximation
  - Curve fitting
  - Numerical differential and integration
  - Approximate solution of differential equations

Assessment of Students

- Students are assessed through:
  1. course work consisting of two in-class tests, test on one Ordinary Differential Equations and Partial Differential Equations and test two on Numerical Methods.
  2. end of semester final examination with Section A of 40 marks and Section B of 60 marks.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Elementary Differential Equations and Boundary Value Problems, Eighth edition by W.E. Boyce and R.C. DiPrima
- Numerical Analysis, 7th Edition by R.L. Burden and Douglas L. Faires
- Partial Differential Equations, 2nd Edition by E. DiBenedetto
- Elements of Partial Differential Equations by P. Drabek and G. Holubova
- Elementary Differential Equations with Boundary Value Problems by H. Edwards, and D. Penney
- Beginning Partial Differential Equations by P.V. O'Neil
- Ordinary Differential Equations by M. Tennenbaum and H. Pollard

Digital Signal Processing

Department of Electronic Engineering

Course Information

Course:	Digital Signal Processing
Course Code:	TEE 3101
Credit Hours:	10

Lecturer Information

Name:	Zedekia Madumbu Nyathi
Email:	zedekia.madumbu.nyathi@nust.ac.zw
Qualifications:	MSc in Electronic Engineering, FETC

Course Synopsis

Analysis of continuous and discrete signals and systems. Fourier series and transforms. Laplace transforms, Z transforms, transfer functions, analysis of stability, probabilistic convolution, impulse response and transfer functions.

Topics

- Analysis of continuous and discrete signals and systems
- Time domain representation of continuous and discrete signals
- Time domain characterisation of linear time invariant continuous and discrete time signals and systems
- Digital Processing of continuous time signals
- Fourier series and transform frequency domain representation of continuous and discrete time signals
- Laplace transforms
- Z transforms
- Transfer functions, derivation of the transfer functions, Analysis of stability

Assessment of Students

- Students are assessed through seven tutorials, assignments, two written in-class tests, all of which shall form the course work assessment mark.
- There shall be final examination at the end of each semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Introductory digital signal processing with computer applications by Paul A. Lynn, Wolfgang Faurst
- Digital signal processing by John G.Proakis,Dimitris G.Manolakis
- Digital signal processing: A Practical Approach by Emmanuel C. Ifeachor, Barrie W.Jervis
- Digital signal processing by Oppenheim A.V and Schafer R.W

Linear Integrated Circuits

Department of Electronic Engineering

Course Information

Course: Linear Integrated Circuits
Course Code: TEE 3111
Credit Hours: 10

Lecturer Information

Name: Svetlana A. Bebova
Email: swetlana.bebova@nust.ac.zw
Qualifications: MSc in Radio Electronic Engineering

Course Synopsis

The module covers operational amplifier circuits: comparators, inverting and non-inverting amplifiers, mathematical operations, oscillators and multivibrators, active filters; Voltage regulators; Timer ICs and their applications; Instrumentation amplifiers; Analogue-to-Digital converters and Digital-to-Analogue converters

Learning Objectives

- To familiarize students with the most commonly used linear integrated circuits and their configurations, properties, characteristics and applications.
- To introduce the analysis methods of various circuits built on operational amplifiers and linear integrated circuits.
- To introduce the design principles for circuits and systems with linear integrated circuits.

Topics

- Operational Amplifiers
- Comparators
- Inverting and Non-Inverting Amplifiers
- Mathematical Operations with Op Amps and Linear Integrated Circuits
- Timers and Signal Generators
- Active Filters
- Voltage Regulators

Assessment of Students

- The continuous assessment of students is through:
  1. two in-class tests, two hours long.
  2. assignments for each topic.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Integrated Circuits Lecture Notes by S. A. Bebova, Electronic Engineering Department, NUST
- Basic Operational Amplifiers and Linear Integrated Circuits, 1994 by T.L. Floyd

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Analogue Communications Engineering

Department of Electronic Engineering

Course Information

Course: Analogue Communications Engineering
Course Code: TEE 3121
Credit Hours: 10

Lecturer Information

Name: Magripa Nleya
Email: magripa.nleya@nust.ac.zw
Qualifications: MSc in Telecommunication Engineering

Course Synopsis

This is an introduction to analogue telecommunications covering amplitude modulation, single-side band modulation, angle modulation, frequency division multiplexing, propagation effects, demodulation. It also covers determination of the signal-to-noise ratio in AM and FM systems. There is design of small-signal HF amplifiers, mixers, oscillators and detectors, mixer theory and spectral analysis, noise generation in electronic circuits and devices.

Learning Objectives

The students will be able to:

- understand telecommunication principles.
- learn about mobile communication systems.
- to have an understanding of modulation and multiplexing methods in telecommunication systems.

Topics

- Communication systems
- Analogue cellular systems
- Amplitude Modulation
- Angle Modulation
- Transmission and propagation
- Multiplexing
- Noise in communication systems

Assessment of Students

- Students are assessed through assignments, laboratory and written in-class tests which shall form the continuous assessment mark.
- There shall be final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Communications engineering principles by Ifiok Otung
- Lecture notes

Software Engineering

Department of Electronic Engineering

Course Information

Course:	Software Engineering
Course Code:	TEE 3132
Credit Hours:	10

Lecturer Information

Name:	Fidelis N. Mugarisanwa
Email:	fidelis.mugarisanwa@nust.ac.zw
Qualifications:	MPhil in Electronic Engineering, BSc in Computer Science

Learning Objectives

The following course objectives provide a general overview of outcome expectations which will depend on the student's motivation and work rate. Thus the following list is just an eye opener. The course will be expected to:

- Introduce the students to the Java programming language, its syntax, and file organisation.
- Introduce students to Java's core libraries.
- Promote the use of the object oriented programming paradigm among the students.
- Demonstrate the potential of the use of Java as a ubiquitous effective problem solving tool.
- Engage students with real world examples.
- Teach students software engineering design concepts.
- Give students practical guidance to:
  - coding and modifying existing code
  - good and bad programming techniques

Topics

- Introduction
- Java Basics Overview
- Using Classes and Objects
- Programming Flow Control Using Decisions
- Iterations
- Methods and Classes
- Arrays
- Inheritance and Polymorphism
- Java Inbuilt Class Examples
- Exceptions

- Threads

Assessment of Students

- Continuous assessment will consist of two written assignments and two written in-class tests.
- The final assessment shall be a written closed book examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Course notes and other materials which may either be supplied or referred to during the course
- Schildt, H. (2015). Java The complete Reference 10th Ed Oracle Press McGraw Hill Education
- Cahoon J. P., Davidson, J. W. (2004). Java 1.5 Design, McGraw Hill Higher Education
- Van der Linden, P. (1996). Just Java SunSoft Press Prentice Hall
- https://www.codejava.net/
- https://www.edureka.co/
- https://www.softwaretestinghelp.com/java/
- http://tutorials.jenkov.com/java/

Electromagnetic Theory

Department of Electronic Engineering

Course Information

Course: Electromagnetic theory
Course Code: TEE 3201
Credit Hours: 10

Lecturer Information

Name: Innocent Muchingami
Email: innocent.muchingami@nust.ac.zw
Qualifications: PhD in Groundwater contaminant transport, MSc in Geophysics, Bsc in Applied Physics Physics

Course Synopsis

The module covers Maxwell's equations i.e. Laplace and Poisson equations and their solution, Boundary conditions, Plane waves in a perfect dielectric, propagation in imperfect dielectric, Propagation in imperfect conductors, skin effect, Generalized wave equation, field distributions in rectangular waveguide, Radiation field, dipoles, radiation resistance, impedance, mutual impedance and linear arrays.

Learning Objectives

- Understand the importance of Vector Analysis in electromagnetism.
- Describe and understand the basic concepts of electrostatics and magneto-statics.
- Master the basic concepts of electrodynamics.
- Understand the laws on conservation as applied to electrodynamics.
- Apply the vector analysis in understanding electromagnetic wave propagation.
- Mathematically solve problems related to electromagnetism using methods of Vector Analysis.

Topics

- Vector Analysis
- Electrostatics
- Electric field in Matter
- Magneto-statics
- Magnetic fields in Matter
- Electrodynamics
- Conservation laws
- Electromagnetic waves

Assessment of Students

Two assignments and two tests as course work. Three hour final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous work (25 %), Final Examination (75 %)

Policies and Other Information

Key Text

- Electromagnetics by J. D. Kraus and K. R. Carver
- Classical electrodynamics by Jackson
- Classical electromagnetic theory by Jack Vanderlinde
- Introduction to electrodynamics by D. J. Griffiths

Digital Communication Engineering

Department of Electronic Engineering

Course Information

Course: Digital Communication Engineering
Course Code: TEE 3221
Credit Hours: 10

Lecturer Information

Name: Magripa Nleya
Email: magripa.nleya@nust.ac.zw
Qualifications: MSc in Telecommunication Engineering

Learning Objectives

The course covers the GSM. The course will make the students understand:

- the principles of digital communications.
- coding.
- modulation and multiplexing methods.

Topics

- Introduction to Digital Communication Systems (DSC)
  - Spectral Density
  - Autocorrelation
  - Random Signals
- Formatting and Baseband Transmission
  - Baseband Systems
  - Formatting textual Data (character coding)
  - Formatting Analog Information
  - Sources of Corruption
  - Pulse Code Modulation
  - Uniform and Non-Uniform Quantization
  - Baseband transmission
- Global System for Mobile Communications(GSM)
  - The GSM subsystems
  - The Network Subsystem
  - The Base Station Subsystem (BSS)
  - Mobility management and call control
  - Call reselection and location area update
  - The mobile-terminated call
  - Handover scenarios

- The mobile device
- The SIM card
- The intelligent network subsystem and CAMEL

Assessment of Students

- Continuous assessment will consist of written assignments and written in-class tests.
- The final assessment shall be an examination written at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Communications Engineering Principles by I. Otung
- Digital Communications: Fundamentals and Applications by B. Sklar
- From GSM to LTE by M. Sauter
- Lecture notes

Embedded Systems

Department of Electronic Engineering

Course Information

Course: Embedded Systems
Course Code: TEE 3232
Credit Hours: 10

Lecturer Information

Name: Reginald Gonye
Email: reginald.gonye@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

The module explores applications of embedded systems in Microcontrollers i.eits memory maps, programming languages, I/O, timers, interrupts, hardware interfacing. It also covers Picocontrollers i.e. memory maps, SFRs, stacks, programming languages, oscillator types, configuration fuses, watchdog timers and code protection.

Learning Aim

To instill concepts of Embedded Computer Systems into students.

Learning Objectives

By the end of the course, candidates should be able to:

- Understand the concept of embedded computers.
- Design an embedded computer solution for a given situation.

Topics

- Introduction
- The 8051 Microcontroller
  1. Introduction
  2. Memory of the 8051
  3. Programming the 8051
  4. IO of the 8051
  5. Counters and timers
  6. Interrupts
- The PIC15F84
  1. Introduction
  2. Memory of the PIC15F84
  3. Programming the PIC15F84
  4. IO of the PIC15F84

5. Counters and timers
6. Interrupts

- Root Locus

Assessment of Students

- Continuous assessment will consist of tests, laboratory work and assignments (NB: Attendance may also contribute to the coursework mark).
- Final assessment will consist of an open book examination at the end of the semester. The examination shall comprise of two sections: section A with three questions on microcontrollers and section B with three questions on PICs. Each question shall carry 33.3 marks. You will be required to answer a total of three questions with at least one from each section. The questions will test your ability to design an embedded computer solution for a given situation.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Lecture notes

Control Engineering

Department of Electronic Engineering

Course Information

Course: Control Engineering
Course Code: TEE 3241
Credit Hours: 10

Lecturer Information

Name: Reginald Gonye
Email: reginald.gonye@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

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Course Synopsis

The module explores examples of controlled processes, objectives and terminology, open and closed-loop controllers. It covers modeling by transfer functions, simple servo mechanisms and derivation of transfer functions from specifications. It also covers time and frequency–response specifications, direct analysis and design, stability, Routh criterion, the ITAE and other performance criteria. There are examples of servo design, frequency-response analysis and design, Root-locus methods and system analysis and design.

Learning Aim

To instill concepts of Control Engineering Theory into students.

Learning Objectives

By the end of the course, candidates should be able to:

- Understand the concept of a control system.
- Understand modelling of control systems.
- Use system models to analyse and synthesise systems.

Topics

- Introduction
- Modeling Control Systems
- Time Response of a System
- Performance Criteria
- Controllers and Compensators
- Stability of a System
- The Root Locus
- Frequency Response
- Introduction to Digital Control

- Application of PLCs in Control Systems
- Case studies of Control Systems

Assessment of Students

- Continuous assessment will consist of tests and assignments (NB: Attendance may also contribute to the coursework mark).
- Final assessment will consist of an examination at the end of the semester. The examination shall comprise of two sections: section A carrying 40 marks and section B with five questions carrying 20 marks each. You will be required to answer all questions in section A and to attempt any three questions from section B. Section A questions will test factual recall of control concepts in form of calculations or accounts. Section B questions will test your ability to apply the concepts learnt in the course to control systems.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Roland S Burns, 2001, Advanced Control Engineering, Butterworth-Heinemann
- D K Anand and R B Zmood, 1995, Introduction to Control Systems Third Edition, Butterworth-Heinemann

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Engineering Management I

Department of Electronic Engineering

Course Information

Course: Engineering Management I
Course Code: TEE 5101
Credit Hours: 10

Lecturer Information

Name: Bhekisisa Nyoni
Email: bhekisisa.dube@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

This module is based on "Management by Engineers" by D. Johnson through group discussion and talks by external speakers. Hence it is centred on industrial organizations, reviews and performance measures, planning and managing change, development and motivating groups, leaderships and communication, financial management, business environment, companies and basic accounts.

Learning Aims

To expose engineering students to managerial concepts and techniques used in the real world by Technical Managers and other technically oriented management personnel. To enable the student to develop coping skills, knowledge and aptitudes for tackling problems of managing creative people.

Learning Outcomes

At the end of the course the student should be able to:

- distinguish between management applied to the technical functions and that applied to operations.
- demonstrate a clear understanding of managerial functions, roles and responsibilities in both technical and operational contexts.
- exhibit knowledge, skills and aptitudes for tackling managerial problems related to leading other professionals.
- conduct in-depth analysis of case studies pertaining to engineering management functions.

Topics

- Introduction to Management for Engineers
  Definitions; employment trends in industry; business environment; future trends, knowledge economies, Personal Strategies for the future and Challenges ahead.
- Functions of Engineering Management (Planning)
  Types of planning; tools for planning; planning activities and strategic management.
- Functions of Engineering Management (Organizing)
  Organisational structure; organising for enhanced corporate performance; delegating and work relationships.

- Functions of Engineering Management (Leading)
  Styles of leadership; decision making; motivating; selecting engineering personnel and developing people.
- Functions of Engineering Management (Controlling
  Types of control; setting performance standards; benchmarking; measuring, evaluating correcting performance.
- Business Essentials (Cost Accounting, Financial Accounting and Management)
  Product or service costing; key financial statements; budgeting forecasting; financial analysis and capital formation.
- Business Essentials (Marketing Management)
  Function of marketing; market forecast; market segmentation; strategies and factors affecting marketing success.
- Operational Excellence
  Tools for Operational Excellence and implementation of Operational Excellence.
- Engineers as Managers/Leaders
  Career path of a typical engineer; Leaders and managers; Leadership styles, qualities attributes and unique contributions expected of engineering managers

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Assessment of Students

- Continuous assessment will consist of in-class presentations, written assignments and written in-class tests.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Management by R. L. Daft
- Modern Project Management by R. C. Mishra, T. Soota
- Change Management by B. Burnes
- Management for Engineers, 1996 by A. C. Payne, J. V. Chelsom, L. R. P. Reavill

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Integrated Circuits and Microelectronics

Department of Electronic Engineering

Course Information

Course: Integrated Circuits and Microelectronics
Course Code: TEE 5111
Credit Hours: 10

Lecturer Information

Name: Svetlana A. Bebova
Email: swetlana.bebova@nust.ac.zw
Qualifications: MSc in Radio Electronic Engineering

Course Synopsis

The module gives a comprehensive view of operational amplifiers i.e. classification, parameters, basic building blocks and data sheets, thermal analysis, applications. It also delves into differential, instrumentation and isolation Amplifiers i.e. its parameters, analysis, data sheets and applications. It also looks into data converters such as ADC, DAC, V/F and F/V converters i.e. its operation, analysis and applications. Elements of data acquisition systems such power supply systems, over-voltage and short circuit protection are covered in this course including microelectronic technologies as well.

Topics

- Operational amplifiers and linear integrated circuits (Ics) classification and parameters.
- Ics basic building blocks
- Microelectronics:
  - Wafer preparation
  - Impurity doping
  - Oxidation
  - Photolithography
  - Metalization
- Integrated components
- Special types amplifiers
  - Differential amplifier (integrated and monolithic)
  - Instrumentation amplifier
  - Isolation amplifier
- Thermal analysis
- Introduction to FPGA
- Data converters
  - Digital-to Analog
  - Analog–to Digital

- – Voltage-to-frequency and Frequency-to-Voltage

Assessment of Students

- Continuous assessment will consist of tests and assignments.
- Final assessment will consist of an examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Communication Systems Performance

Department of Electronic Engineering

Course Information

Course: Communication Systems Performance
Course Code: TEE 5121
Credit Hours: 10

Lecturer Information

Name: Zedekia Madumbu Nyathi
Email: zedekia.madumbu.nyathi@nust.ac.zw
Qualifications: MSc in Electronic Engineering, FETC

Course Synopsis

This course delves into design and performance analysis of communication circuits i.e. tuned amplifiers, impedance matching, oscillators, filters, transmission lines and phase locked loops. Then it covers some application examples of the circuits.

Topics

- The concepts of Noise Characterisation and Receiver Performance
- Overview of Contemporary Communication Systems Link Budgets
- Introduction to Random processes and Spectral Analysis
- Random processes and Spectral analysis
  - Linear Systems
  - Bandwidth Measurement
  - The Gaussian Random Process
  - Bandpass Processes
  - Matched filters
- Performance of Communication Systems Corrupted by Noise Error Probabilities for Binary Signalling
- Performance of Communication Systems Corrupted by Noise
  - Performance of Baseband Binary Systems
  - Detection of Bandpass Binary Signals
  - Comparison of Digital Signalling Systems

Assessment of Students

- Continuous assessment will consist of written in-class tests and assignments.
- Final assessment will consist of an examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Digital and Analog Communication Systems, 1993 by L. W. Couch II.
- Digital Communications, 1988 by B. Sklar
- Digital Communications, 1995 by J. G. Proakis
- Digital and Analog Communication Systems, 1994 by K. S. Shanmugan
- Communication Systems by A. B. Carlson
- Communication Systems, 1994 by S. Haykin

Computer Architecture and Operating Systems

Department of Electronic Engineering

Course Information

Course:	Computer Architecture and Operating Systems
Course Code:	TEE 5133
Credit Hours:	10

Lecturer Information

Name:	Nomakhosi Ndiweni
Email:	nomakhosi.ndiweni@nust.ac.zw
Qualifications:	PhD in Electronic Engineering, MEng in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

The module examines the evolution of computer hardware for Von Neumann machines. It also covers operating systems for single tasking, process scheduling for concurrent operation, inter-process communication, deadlock avoidance and memory management. The module also delves into virtual memory, architectures for parallel processing and computer networking.

Topics

- The Evolution of Computer Systems
- Hardware Architectures for Von Neumann Machines
- Operating Systems for Single-Tasking Computers
- Process Scheduling in Concurrent Operating Systems
- Inter-Process Communication
- Deadlock Avoidance in Multiprogrammed Computers
- Memory Management in Multiprogrammed Machines
- Hardware Architectures for Parallel Processing
- Computer Networking

Assessment of Students

- Continuous assessment will consist of research and presentations, assignments and written in-class tests.
- Final assessment will consist of an examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Operating Systems: Design and Implementation, 3rd Edition by A. Tanebaum, A. Woodhull

Modern Control Engineering

Department of Electronic Engineering

Course Information

Course: Modern Control Engineering
Course Code: TEE 5141
Credit Hours: 10

Lecturer Information

Name: Reginald Gonye
Email: reginald.gonye@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

The module looks at State Space Analysis covering State-space methods of analysis and design, Observability and controllability, Pole placement for the optimization response, State observers and pole placement design with state observers, Multi-input, multi-output systems and cross-coupling problems. It also delves into Digital Control covering Digital time control systems, Modeling of Sampled Processes, Transient response, Steady state response, Stability, Design of Digital Controllers and Root Locus.

Learning Aim

To instill state space analysis methods for Control Engineering into students. To instill discrete analysis methods for Control Engineering into students.

Learning Objectives

By the end of the course, candidates should be able to:

- Understand the concept of states as applied to Control Systems.
- Be able to use state space analysis in the analysis of Control Systems.
- Understand the concept of discrete analysis as applied to digital control systems.
- Be able to analyse digital control systems.

Topics

- A review of Control Engineering
- Models of Systems
- State Space Representations
- Time Response from State Space Equations
- State Space Realizations
- State Feedback
- Review of analysis of Discrete Signal
- Modelling of Sampled Processes

- Performance e Features of Digital Control Systems
- Design of Digital Controllers
- Root Locus

Assessment of Students

- Continuous assessment will consist of tests and assignments (NB: Attendance may also contribute to the coursework mark).
- Final assessment will consist of an examination at the end of the semester. The examination shall comprise of six questions each carrying 25 marks. You will be required to attempt four of the six questions.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Advanced Control Engineering by R. S. Burns
- Introduction to Control Systems by D. K. Anand, R. B. Zmood
- Control Engineering by E. A. Parr

Engineering Management II

Department of Electronic Engineering

Course Information

Course: Engineering Management II
Course Code: TEE 5205
Credit Hours: 10

Lecturer Information

Name: Bhekisisa Nyoni
Email: bhekisisa.dube@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Introduction

Project management is an essential skill-set for many careers and in many contexts in our lives. Project Management is an ideal starting point if you need to manage projects at work or at home, while not necessarily being a formally trained project manager. It is also suitable if you are considering undertaking a project in the near future and are seeking to learn and apply essential project management knowledge and skills.

Learning Aims

The course seeks to acquaint the learners with the principles of managing projects successfully. It also seeks to help them integrate these processes into their daily lives and particularly the successful undertaking of their Final year projects.

Learning Outcomes

At the end of the course the student should be able to:

- Understand the growing need for better project management in industry.
- Explain a project is and provide examples of projects.
- Describe what project management and discuss key elements of the project management framework.
- Discuss how project management relates to other management discipline.
- Understand the project management lifecycle and be knowledgeable on the various phases from project initiation through closure.
- Utilize the use of a structured approach for each and every unique project undertaken including utilizing project management concepts, tools, and techniques.
- Develop detailed project plan to include: Defining a project's scope and tasks, estimating activity durations, estimating task resource needs, assessing project risk and response strategies, a communications plan, and more.
- Integrate project management concepts into doing their Final year projects.

Topics

- Introduction to Project Management
- Project Life cycle and System Approach
- Project Scoping
- Project Time Estimation and Scheduling
- Project Costing
- Project Quality Management
- Project Communication Recourse, Stakeholder and Procurement Management
- Project Monitoring and Evaluation
- Agile Project Management and Introduction to Project Management Information Systems

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Assessment of Students

- Continuous assessment will consist of in-class presentations, written assignments and written in-class tests.
- There shall be a final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- A Guide to the Project Management Body of Knowledge, 6th Edition by Project Management Institute
- Successful Project Management, 4th Edition, 2009 by J. Gido and J. P. Clements
- Project Management for Business, Engineering, and Technology, 3rd Edition, 2008 by J. M. Nicholas, H. Steyn

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High Speed Networks

Department of Electronic Engineering

Course Information

Course: High Speed Networks
Course Code: TEE 5224
Credit Hours: 10

Lecturer Information

Name: Zedekia Madumbu Nyathi
Email: zedekia.madumbu.nyathi@nust.ac.zw
Qualifications: MSc in Electronic Engineering, FETC

Course Synopsis

A comprehensive view of high-speed LAN, MAN and ATM technologies and standards. Evolution towards broadband integrated services digital network.

Topics

- Data Communications: Introduction, Parallel Transmission, Serial Transmission
- Data Communication Standards
- Reference Models and Layered Protocols
- High Speed Local Area Networks
- High Speed Communication Links
- Performance Evaluation of Communication Networks

Assessment of Students

- Students are assessed through two written in-class tests which shall form the continuous assessment mark.
- There shall be final examination at the end of the semester.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)

Policies and Other Information

Key Text

- Computer Networks, 5th Edition by Larry L. Peterson and Bruce S. Davie
- Data and Computer Communications, 5th Edition by W. Stallings
- Computer Networks, 3rd Edition by A.S. Tanenbaum
- Performance Evaluation of Communication Networks, 1st Edition by G. N. Higginbottom

- Absolute Beginner's Guide to Networking, 2nd Edition by M. Gibbs, T. Brown

Course Information

Course: Industrial Control
Course Code: TEE 5241
Credit Hours: 10

Lecturer Information

Name: Reginald Gonye
Email: reginald.gonye@nust.ac.zw
Qualifications: MPhil in Electronic Engineering, (Hon) BEng in Electronic Engineering

Course Synopsis

The module focuses on industrial control situations, process control, instrumentation, actuators, transducers and controllers, hybrid systems, time-domain analysis, state-space analysis, stability; computer control, system characterization, algorithm design, feedback control for digital systems and PLC applications.

Learning Aim

To introduce students to practical arrangements of Control Systems in industry. To instill skills of developing PLC based solutions for control situations into students

Learning Objectives

By the end of the course, candidates should be able to:

- Understand the arrangement of typical control systems in industry.
- Understand the operation of a PLC.
- Be able to develop PLC-based solutions for various control situations.

Topics

- A review of Control Systems
- Sensors and Transducers
- Actuators and their Control
- Computers in Industrial Control
- The PLC
- SCADA

Assessment of Students

- Continuous assessment will consist of tests and assignments (NB: Attendance may also contribute to the coursework mark).

- Final assessment will consist of an examination at the end of the semester. The examination shall comprise of six questions each carrying 25 marks. You will be required to attempt four of the six questions.

Course Evaluation and Grading Policies

Grades are based on: Continuous assessment (25 %), Final Exam (75 %)